Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Experiment:-1

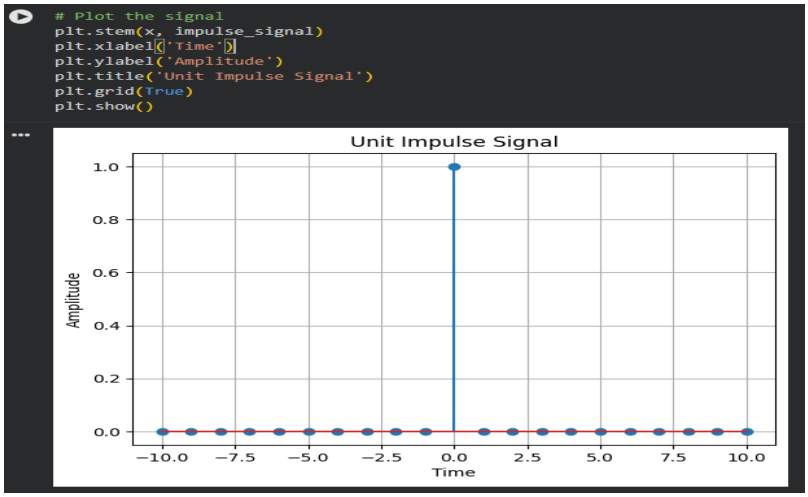
GITHUB: https://github.com/Chiragsinh933/DSIP.git

AIM:- Simulate Discrete time Sequences

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
def unit_impulse(length, position):
    signal = np.zeros(length)
    signal[position] = 1
    return signal
# Parameters
start = -10 # Start value of the x-axis range
stop = 10 # Stop value of the x-axis range
step = 1 # Step size
# Generate x-axis values
x = np.arange(start, stop+step, step)
# Generate unit impulse signal
impulse_signal = unit_impulse(len(x), abs(start)//step)
# Plot the signal
plt.stem(x, impulse_signal)
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.title('Unit Impulse Signal')
plt.grid(True)
plt.show()
```

Output:-



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Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

AIM:- Simulate impulse train

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
def simulate_impulse_train(signal_length, period):
 impulse_train = np.zeros(signal_length)
 for n in range(signal_length):
 if n % period == 0:
 impulse_train[n] = 1
 return impulse_train
# Define the parameters for the impulse train
signal_length = 100 # Length of the impulse train
period = 10 # Period of the impulse train
# Simulate the impulse train
impulse_train = simulate_impulse_train(signal_length, period)
# Plot and display the impulse train
plt.stem(impulse_train)
plt.title('Impulse Train')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.show()
```

output:-

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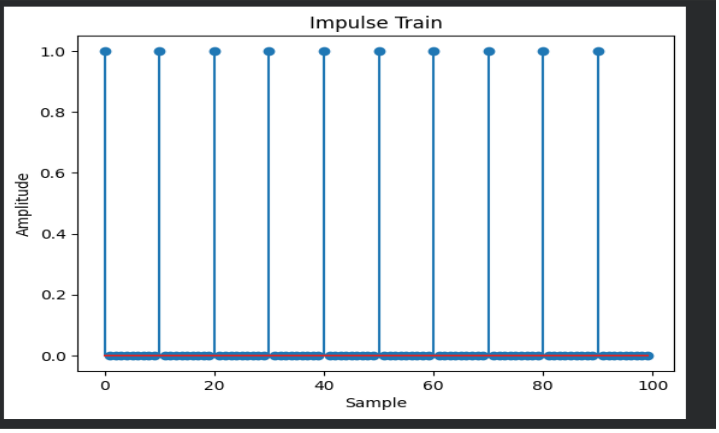
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

```
plt.title('Impulse Train')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.show()
```



AIM:Write a python program to Simulate continuous and discrete unit step signal.

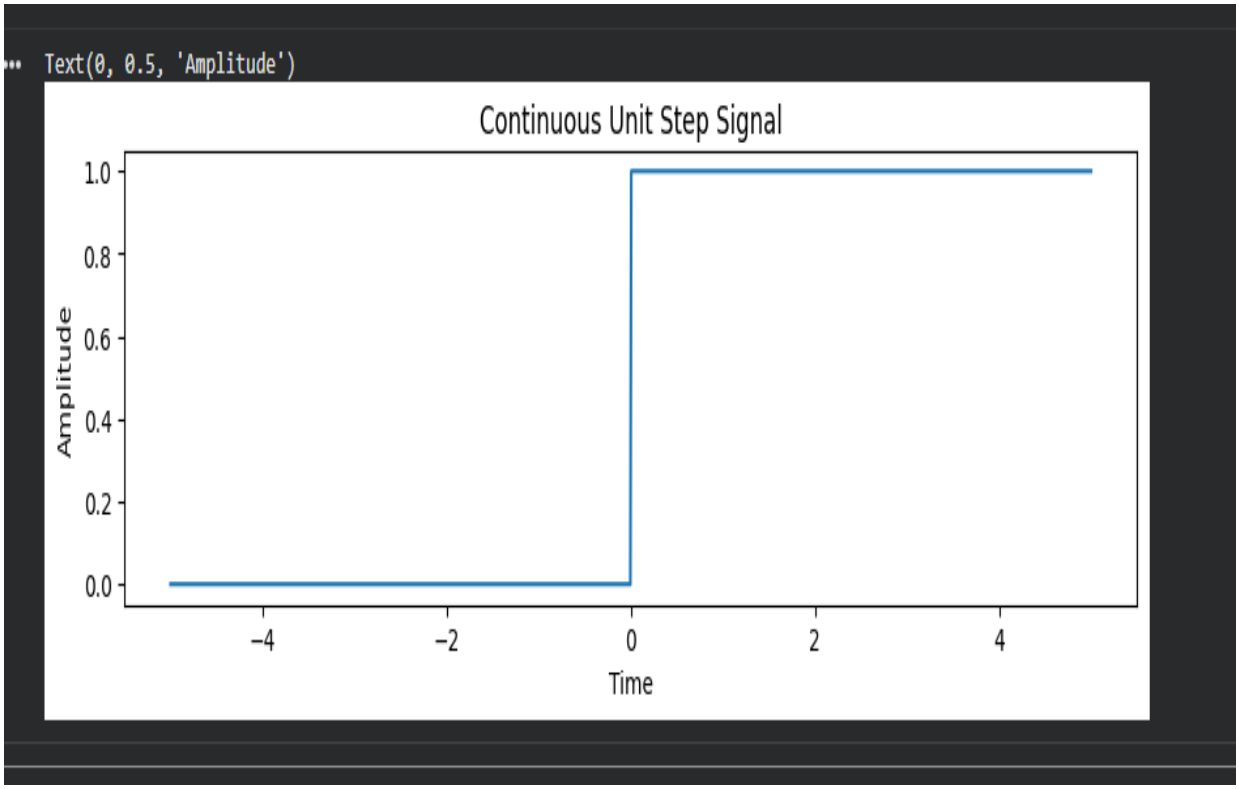
code:-

```
import numpy as np
import matplotlib.pyplot as plt
def simulate_continuous_exponential(time, amplitude, coefficient):
 exponential_signal = amplitude * np.exp(coefficient * time)
 return exponential_signal
def simulate_discrete_exponential(num_samples, amplitude, coefficient):  exponential_signal =
amplitude * np.exp(coefficient * np.arange(num_samples))  return exponential_signal
# Define the time range for the continuous exponential signal
time = np.linspace(0, 5, 1000) # Time range from 0 to 5
# Define the number of samples, initial amplitude, and coefficient for the discrete
exponential  signal
num_samples = 20 # Number of samples
amplitude = 2 # Initial amplitude
coefficient = -0.5 # Exponential coefficient
# Simulate the continuous exponential signal
continuous_exponential = simulate_continuous_exponential(time, amplitude, coefficient)
# Simulate the discrete exponential signal
discrete_exponential = simulate_discrete_exponential(num_samples, amplitude, coefficient)
# Plot and display the continuous and discrete exponential signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

```
plt.plot(time, continuous_exponential)
plt.title('Continuous Exponential Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_exponential)
plt.title('Discrete Exponential Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

output:-



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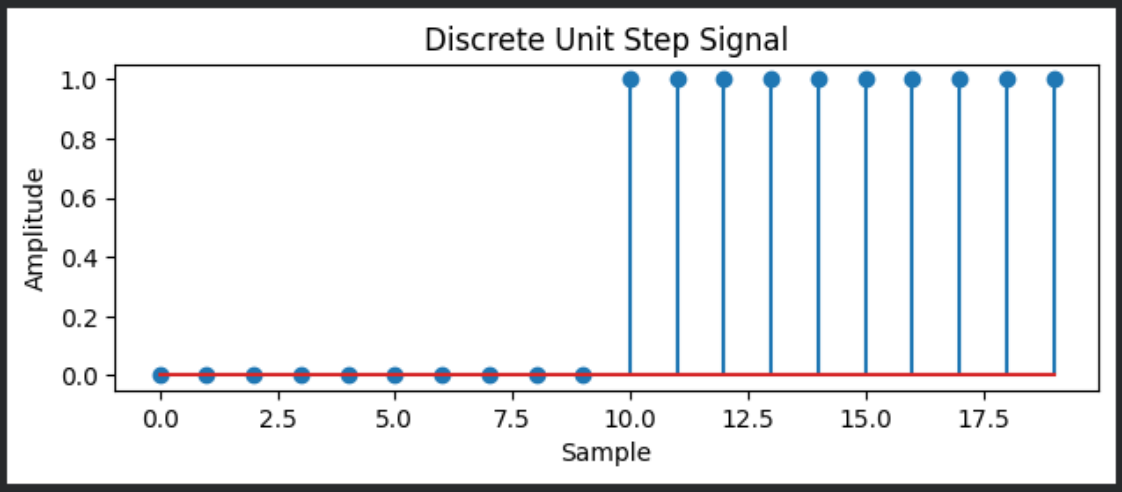
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

```
# np.savetxt('discrete_unit_step.txt', discrete_unit_step, delimiter=',')
```



AIM:- Write a python program to Simulate continuous and discrete ramp signal.

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
def simulate_continuous_ramp(time, slope):
  ramp = np.zeros_like(time)
  ramp[time >= 0] = slope * time[time >= 0]
  return ramp
def simulate_discrete_ramp(num_samples, slope):
 ramp = np.zeros(num_samples)
 ramp[num_samples // 2:] = slope * np.arange(num_samples // 2, num_samples)
 return ramp
# Define the time range for the continuous ramp signal
time = np.linspace(-5, 5, 1000) # Time range from -5 to 5
# Define the number of samples and slope for the discrete ramp signal
num_samples = 20 # Number of samples
slope = 2 # Slope of the ramp
# Simulate the continuous ramp signal
continuous_ramp = simulate_continuous_ramp(time, slope)
```

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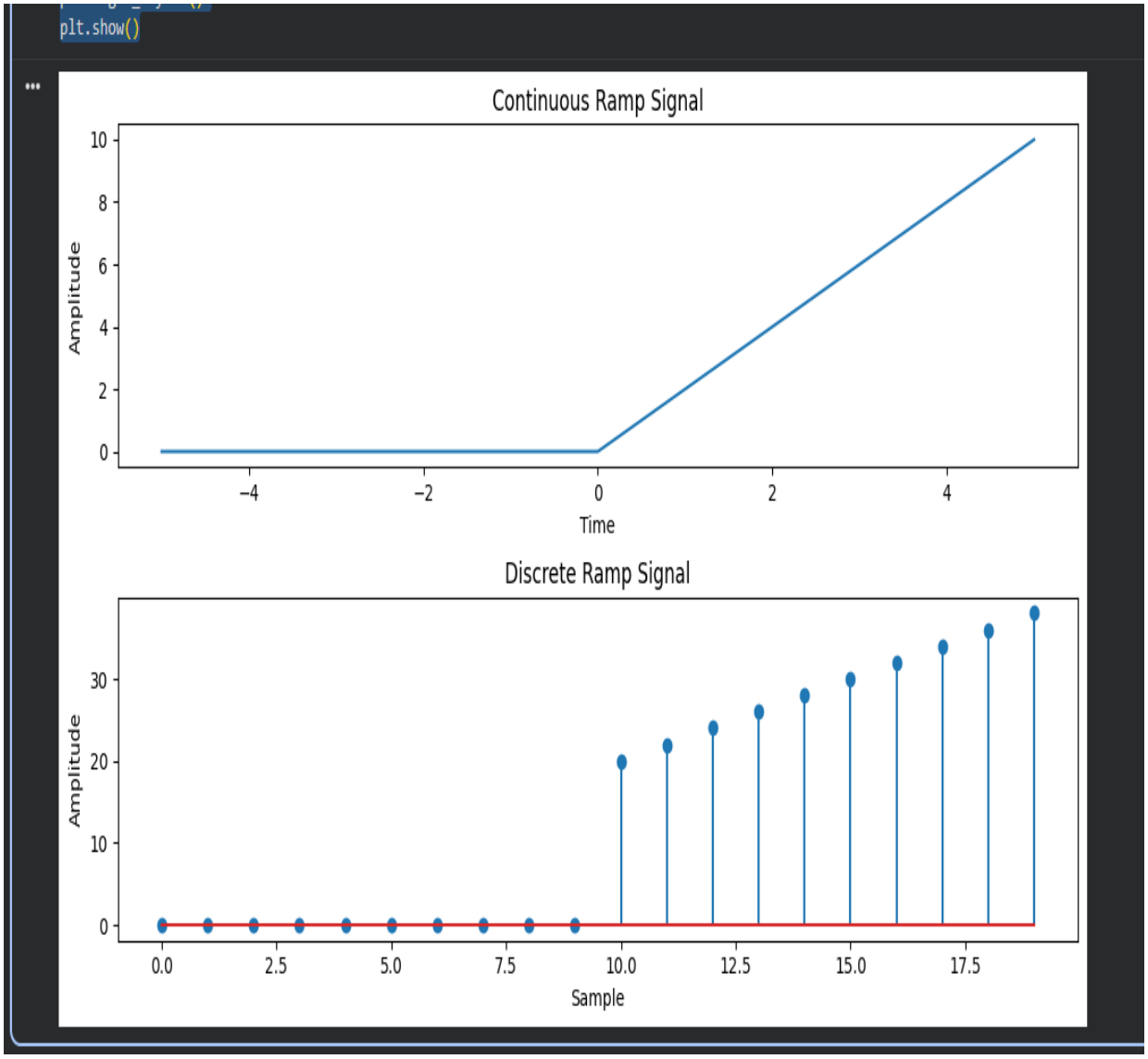
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

```
# Simulate the discrete ramp signal
discrete_ramp = simulate_discrete_ramp(num_samples, slope)
# Plot and display the continuous and discrete ramp signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_ramp)
plt.title('Continuous Ramp Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_ramp)
plt.title('Discrete Ramp Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

output:-

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**



AIM:- Write a python program to Simulate continuous and discrete exponential signal.

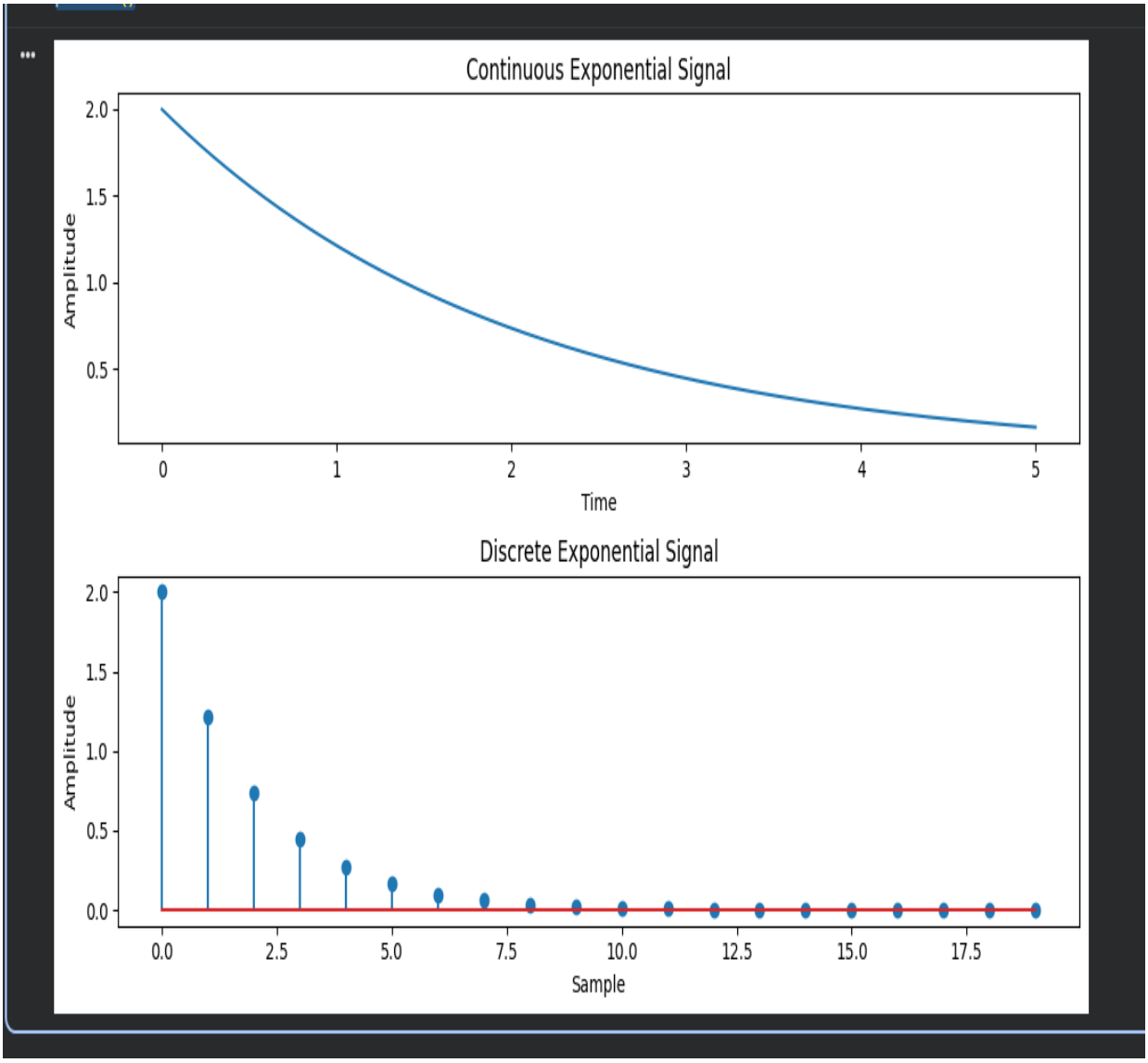
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Code:-

```
def simulate_continuous_exponential(time, amplitude, coefficient):
 exponential_signal = amplitude * np.exp(coefficient * time)
 return exponential_signal
def simulate_discrete_exponential(num_samples, amplitude, coefficient):
 exponential_signal = amplitude * np.exp(coefficient * np.arange(num_samples))
 return exponential_signal
# Define the time range for the continuous exponential signal
time = np.linspace(0, 5, 1000) # Time range from 0 to 5
# Define the number of samples, initial amplitude, and coefficient for the discrete exponential
# signal
num_samples = 20 # Number of samples
amplitude = 2 # Initial amplitude
coefficient = -0.5 # Exponential coefficient
# Simulate the continuous exponential signal
continuous_exponential = simulate_continuous_exponential(time, amplitude, coefficient)
# Simulate the discrete exponential signal
discrete_exponential = simulate_discrete_exponential(num_samples, amplitude, coefficient)
# Plot and display the continuous and discrete exponential signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_exponential)
plt.title('Continuous Exponential Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_exponential)
plt.title('Discrete Exponential Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

output:-

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**



AIM:- Write a python program to Simulate continuous and discrete parabolic signal.

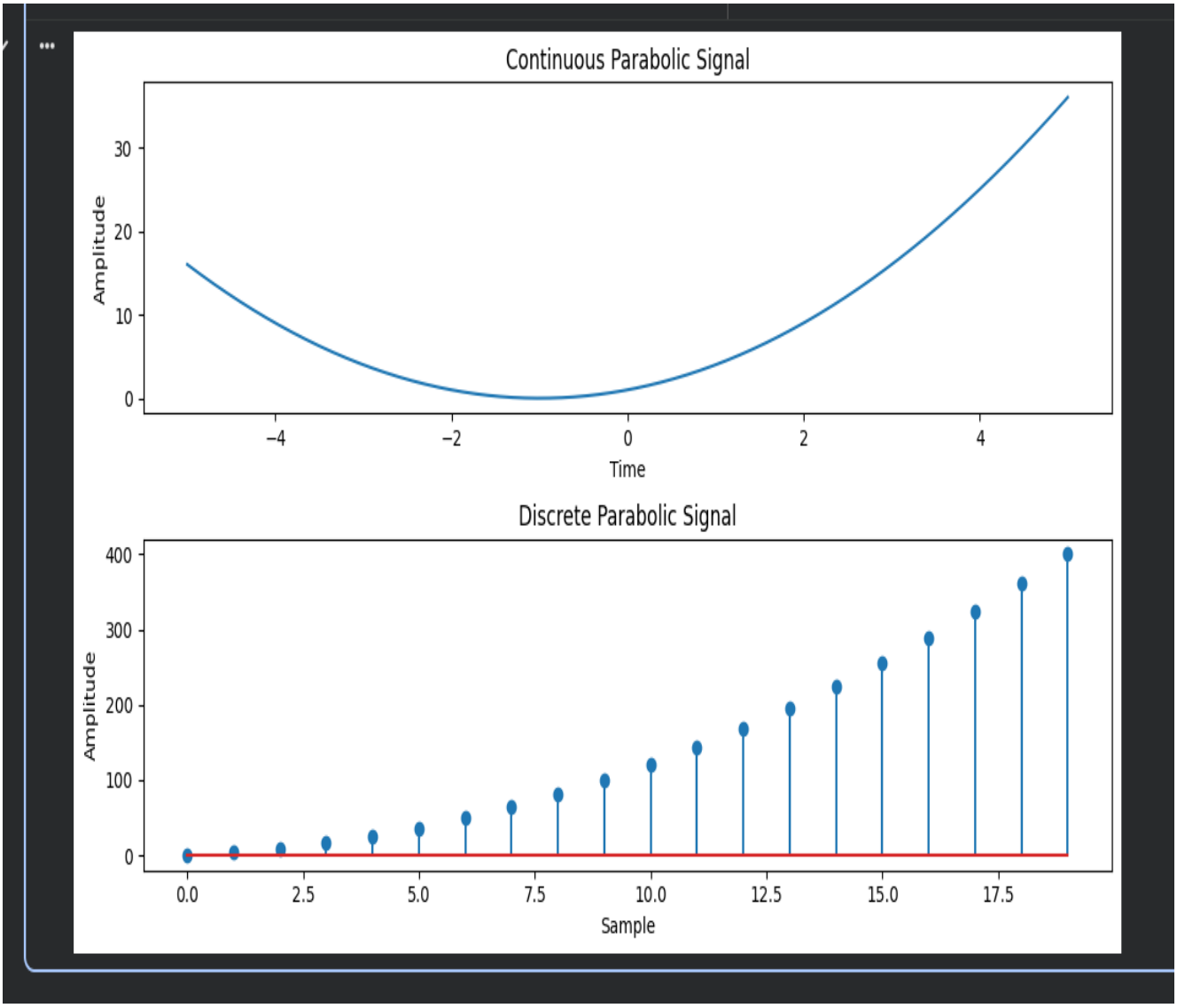
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Code:-

```
def continuous_parabolic(time, coefficients):
 parabolic_signal = np.polyval(coefficients, time)
 return parabolic_signal
def simulate_discrete_parabolic(num_samples, coefficients):
 parabolic_signal = np.polyval(coefficients, np.arange(num_samples))
 return parabolic_signal
# Define the time range for the continuous parabolic signal
time = np.linspace(-5, 5, 1000) # Time range from -5 to 5
# Define the number of samples and coefficients for the discrete parabolic signal
num_samples = 20 # Number of samples
coefficients = [1, 2, 1] # Coefficients of the parabolic signal
# Simulate the continuous parabolic signal
continuous_parabolic = continuous_parabolic(time, coefficients)
# Simulate the discrete parabolic signal
discrete_parabolic = simulate_discrete_parabolic(num_samples, coefficients)
# Plot and display the continuous and discrete parabolic signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_parabolic)
plt.title('Continuous Parabolic Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_parabolic)
plt.title('Discrete Parabolic Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

output:-

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**



AIM:- Write a python program to Simulate continuous and discrete sine wave signal.

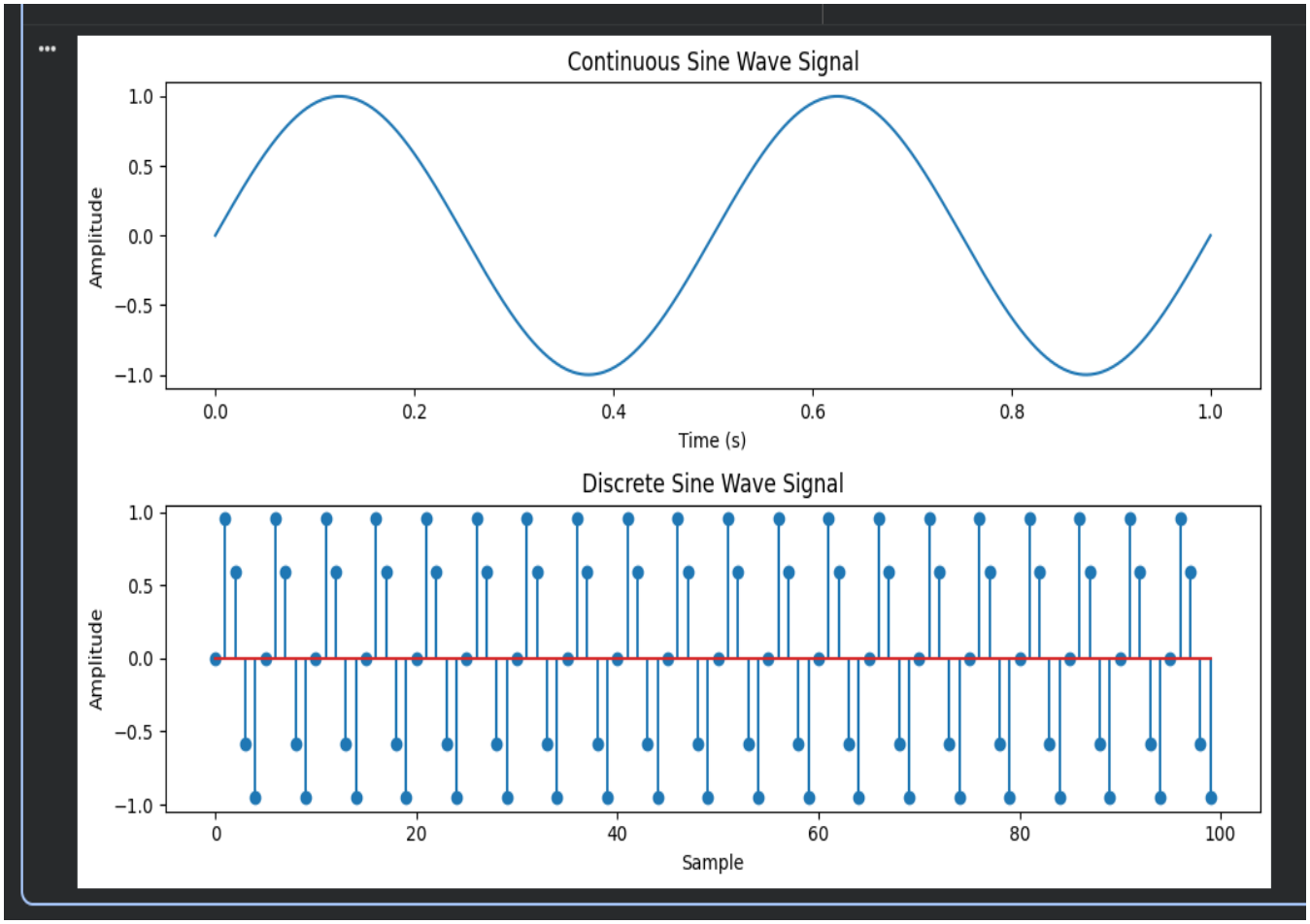
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Code:-

```
def simulate_continuous_sine_wave(time, amplitude, frequency, phase):
 sine_wave = amplitude * np.sin(2 * np.pi * frequency * time + phase)
 return sine_wave
def simulate_discrete_sine_wave(num_samples, sampling_frequency, amplitude, frequency,
phase):
 time = np.arange(num_samples) / sampling_frequency
 sine_wave = amplitude * np.sin(2 * np.pi * frequency * time + phase)
 return sine_wave
# Define the time range for the continuous sine wave signal
time = np.linspace(0, 1, 1000) # Time range from 0 to 1 second
# Define the number of samples, sampling frequency, and parameters for the discrete sine wave
signal
num_samples = 100 # Number of samples
sampling_frequency = 10 # Sampling frequency in Hz
amplitude = 1 # Amplitude of the sine wave
frequency = 2 # Frequency of the sine wave in Hz
phase = 0 # Phase angle of the sine wave in radians
# Simulate the continuous sine wave signal
continuous_sine_wave = simulate_continuous_sine_wave(time, amplitude, frequency, phase)
# Simulate the discrete sine wave signal
discrete_sine_wave = simulate_discrete_sine_wave(num_samples, sampling_frequency,
amplitude, frequency, phase)
# Plot and display the continuous and discrete sine wave signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_sine_wave)
plt.title('Continuous Sine Wave Signal')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_sine_wave)
plt.title('Discrete Sine Wave Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

output:-

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**



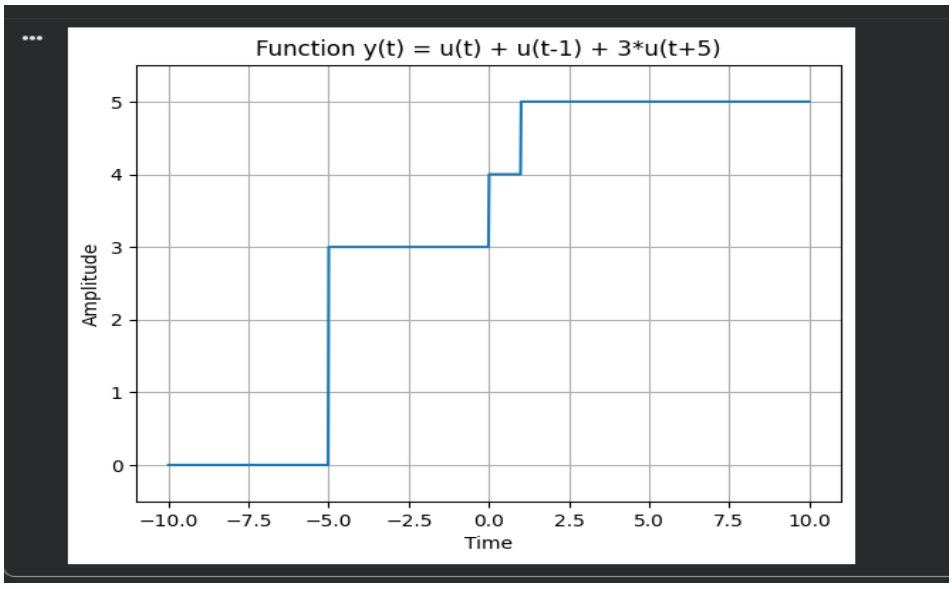
AIM:- Write a python program to simulate y(t)=u(t)+u(t-1)+3u(t+5).

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Code:-

```
def simulate_function(time):
 y = np.zeros_like(time)
 y[time >= 0] = 1
 y[time >= 1] += 1
 y[time >= -5] += 3
 return y
# Define the time range
time = np.linspace(-10, 10, 1000)
# Simulate the function
function_values = simulate_function(time)
# Plot and display the function
plt.plot(time, function_values)
plt.title('Function y(t) = u(t) + u(t-1) + 3*u(t+5)')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.ylim([-0.5, 5.5])
plt.grid(True)
plt.show()
```

output:-



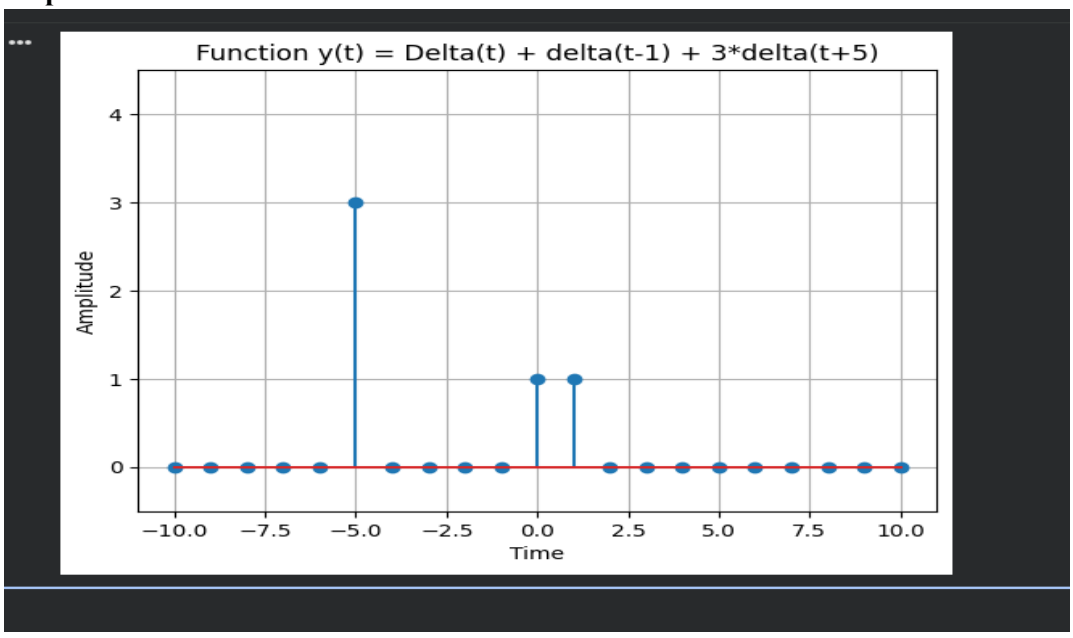
AIM:-Write a python program to simulate y(t)=Delta(t)+delta(t-1)+3*delta(t+5).

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Code:-

```
def simulate_function(time):
 y = np.zeros_like(time)
 y[time == 0] = 1
 y[time == 1] += 1
 y[time == -5] += 3
 return y
# Define the time range
time = np.arange(-10, 11)
# Simulate the function
function_values = simulate_function(time)
# Plot and display the function
plt.stem(time, function_values)
plt.title('Function y(t) = Delta(t) + delta(t-1) + 3*delta(t+5)')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.ylim([-0.5, 4.5])
plt.grid(True)
plt.show()
```

output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:10-07-2026**	**Enrollment No:92400133075**

Conclusion:-

Signal Type	Continuous Form	Discrete Form	Key Parameter
Unit Impulse	δ(t)	δ[n]	Impulse at n = 0
Impulse Train	Periodic δ(t)	Periodic δ[n]	Period
Unit Step	u(t)	u[n]	Step at t = 0
Ramp	r(t) = a·t	r[n] = a·n	Slope (a)
Exponential	A·e^(bt)	A·e^(bn)	Exponential coefficient (b)
Parabolic	$at^2 + bt + c$	$an^2 + bn + c$	Coefficients (a, b, c)
Sine Wave	A·sin(2πft + φ)	A·sin(2πfn/Fs + φ)	Frequency (f), Phase (φ)
Composite Signals	Combination of basic signals	Combination of basic signals	Multiple signal components

Using modular functions makes the implementation simple, reusable, and easy to modify for different signal parameters. The program can efficiently generate various signal types, including impulse, step, ramp, exponential, parabolic, sine wave, and composite signals. This approach provides a strong foundation for studying advanced DSP concepts, communication systems, signal analysis, and control engineering while improving the understanding of the relationship between continuous and discrete signal representations.